Pharma MES Prototype: Queueing Theory, Discrete-Event Simulation, and MES Integration for a Sterile Injectable Manufacturing Facility

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This paper presents a fully integrated Manufacturing Execution System (MES) prototype for a sterile injectable pharmaceutical facility. The system combines analytical queueing theory, a SimPy discrete-event simulation, rigorous statistical analysis, and a PostgreSQL-backed production planning pipeline — all delivered through a live web-based dashboard. Five key concepts are demonstrated: queueing models (M/M/1, M/M/c, M/G/1), a nine-station DES factory model, 15-replication statistical analysis with confidence intervals and paired t-tests, five-scenario what-if analysis, and an ISA-95 Level 3 MES integration covering MRP explosion, EDD/SPT scheduling, and linear programming optimisation. The bottleneck is identified as Aseptic Filling ($\rho = 0.50$); adding a second filling machine reduces queue wait by 93% and is the single highest-impact improvement available.

Introduction

Pharmaceutical manufacturing of sterile injectable drugs is among the most tightly regulated and operationally complex segments of the process industry. Every batch must pass through a fixed sequence of validated steps — raw material staging, quarantine hold, dispensing, compounding, sterile filtration, aseptic filling, inspection, and quality release — before it can be distributed. Delays, equipment failures, or scheduling inefficiencies at any station ripple forward through the entire process, directly impacting patient supply and regulatory compliance.

This project develops a **Pharma MES Prototype** that models such a facility at the system level. The prototype addresses a central operations question: *given a fixed production process and a stream of incoming batches, where does congestion develop, how severe is it, and what interventions produce the greatest improvement?*

The system was designed to demonstrate five manufacturing engineering concepts within a single cohesive platform:

1. **Queueing Theory** — analytical prediction of steady-state waiting times at each station.
2. **Discrete-Event Simulation** — a SimPy-based virtual factory that replicates batch flow with stochastic service times and machine breakdowns.
3. **Statistical Analysis** — 15-replication experiments with Welch warm-up, Student-t confidence intervals, and paired t-tests using Common Random Numbers.
4. **What-If Scenario Analysis** — five structural and operational modifications tested against

the baseline before any real-world commitment.

5. **MES Integration** — a PostgreSQL production database covering Material Requirements Planning (MRP), EDD/SPT scheduling, PuLP linear programme optimisation, and an ISA-95 Level 3 data flow.

All five components are accessible from a single web dashboard running at `http://localhost:8090`, launched by a one-command Python script (`RUN_PROJECT.py`). The dashboard features a live animated factory floor, real-time analytics charts, a full simulation results tab, and live MRP/scheduling tables drawn directly from the database.

System Architecture and How It Works

Factory Process

The simulated facility produces three sterile injectable products — Injectable-A (100 mg/mL), Injectable-B (50 mg/mL), and Injectable-C (25 mg/mL) — in 10 mL glass vials. Every batch follows an identical nine-station sequence shown in Table 1.

Table 1: Factory Process — Station Sequence and Processing Times

Step	Station	Service Time	Notes
(1)	Raw Material Staging	∼30 min (Lognormal)	
(2)	Quarantine Hold	120 min (fixed)	Regulatory dwell
(3)	Dispensing & Weighing	∼45 min (Lognormal)	
(4)	Compounding / Solution Prep	∼90 min (Lognormal)	
(5)	Sterile Filtration	∼60 min (Lognormal)	0.22 μm filter
(6)	**Aseptic Filling**	**∼120 min (Lognormal)**	**Bottleneck**
(7)	Inspection & Packaging	∼45 min (Lognormal)	
(8)	QA Release Hold	180 min (fixed)	Regulatory dwell
(9)	Released	—	

Batches arrive according to a Poisson process with mean inter-arrival time of 240 minutes ($\lambda = 1/240$ batches/min). Product type is assigned randomly: 40% Injectable-A, 35% Injectable-B, 25% Injectable-C, with processing time multipliers of 1.00, 0.88, and 0.74 respectively.

Two Simulation Modes

The prototype operates two distinct simulation engines. The **statistical simulation** (`des_simulation.py`) runs 15 complete replications of 15,000 simulated minutes (≈10 days each) across five scenarios and produces statistically reliable results. The **live dashboard simulation** runs continuously in the browser, displaying individual batch tokens moving through a real-time SVG factory diagram — it serves as a visualisation and teaching tool rather than a source of statistical conclusions.

Software Stack

The system is implemented entirely in Python with no web framework (pure `ThreadingHTTPServer`). SimPy 4 provides the discrete-event engine; psycopg2 connects to a PostgreSQL 14 database (`Sheridan_573`); Chart.js renders the browser charts. The entire system launches from a single file: `python3 RUN_PROJECT.py`.

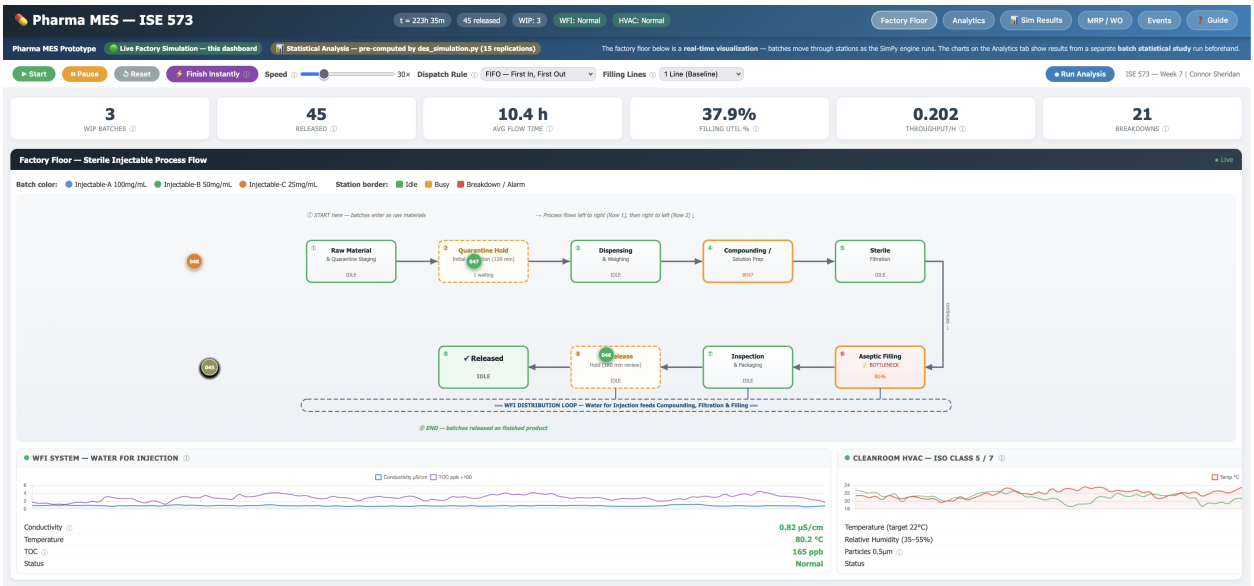


Figure 1: MES Dashboard — Factory Floor tab. Coloured batch tokens (blue = Injectable-A, green = Injectable-B, orange = Injectable-C) move through the nine-station process diagram in real time. Station borders indicate idle (green), busy (orange), or breakdown/alarm (red). Live WFI and HVAC charts are displayed below the process diagram.

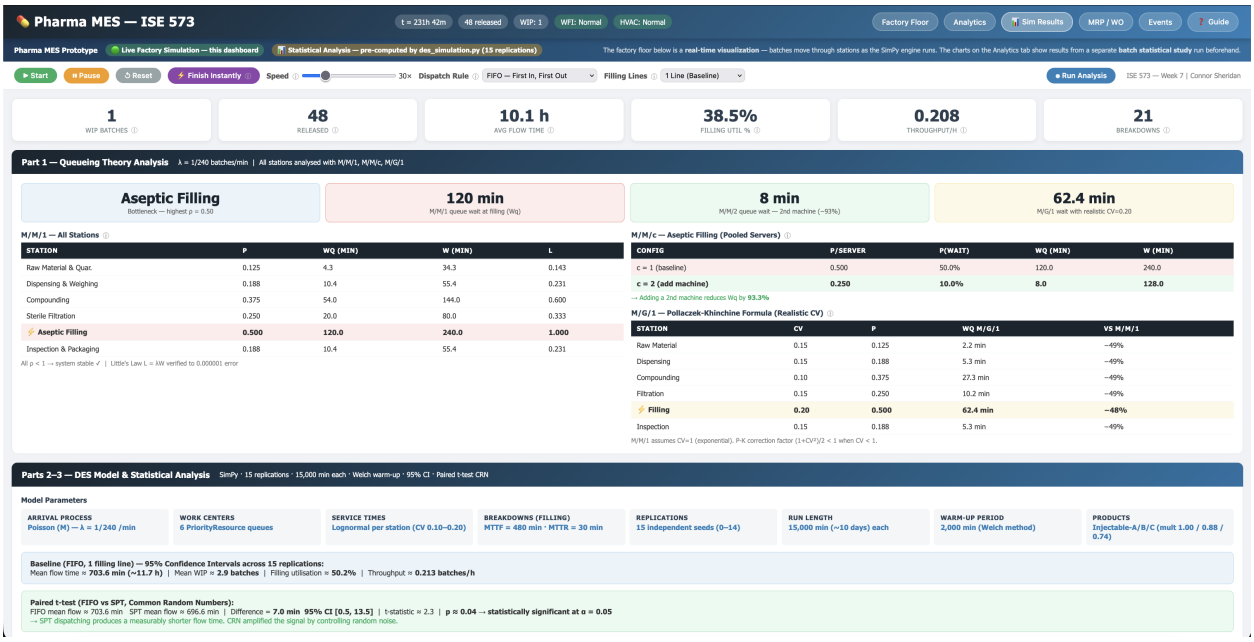


Figure 2: MES Dashboard — Sim Results tab, showing the queueing theory summary (Part 1, top) and DES model parameters with baseline 95% confidence intervals and paired t-test results (Parts 2–3, bottom). All values are drawn live from the `simulation_results` database table.

Part 1 — Queueing Theory Analysis

Before running any simulation, each station was modelled analytically to establish theoretical steady-state predictions. Three queueing models were applied.

M/M/1 — All Stations

Assuming Poisson arrivals, exponential service, and a single server at each station, the standard M/M/1 formulas give the results in Table 2. All utilisation values satisfy $\rho < 1$, confirming system stability. Little's Law ($L = \lambda W$) was verified to a maximum error of 6×10^{-7} across all stations.

Table 2: M/M/1 Queueing Results — All Variable Stations ($\lambda = 1/240\,\text{min}^{-1}$)

Station	$\boldsymbol{\rho}$	$\boldsymbol{W_q}$ (min)	$\boldsymbol{W}$ (min)	$\boldsymbol{L}$
Raw Material & Quarantine	0.125	4.3	34.3	0.143
Dispensing & Weighing	0.188	10.4	55.4	0.231
Compounding	0.375	54.0	144.0	0.600
Sterile Filtration	0.250	20.0	80.0	0.333
Aseptic Filling	**0.500**	**120.0**	**240.0**	**1.000**
Inspection & Packaging	0.188	10.4	55.4	0.231

Aseptic Filling carries the highest utilisation ($\rho = 0.50$) and the longest queue wait ($W_q = 120\,\text{min}$), identifying it unambiguously as the system bottleneck.

M/M/c — Pooled Servers at Aseptic Filling

Using the Erlang-C formula for $c = 2$ parallel filling machines:

Table 3: M/M/c Results — Aseptic Filling, 1 vs. 2 Machines

Configuration	ρ/server	P(wait)	W_q (min)	W (min)
$c = 1$ (baseline)	0.500	50.0%	120.0	240.0
$c = 2$ (add machine)	0.250	10.0%	**8.0**	128.0

Adding a second filling machine reduces W_q from 120 min to 8 min, a **93.3% reduction**. The probability of any arriving batch having to wait drops from 50% to 10%. This is the most impactful single improvement available to the system.

M/G/1 — Realistic Service Variability

The M/G/1 Pollaczek-Khinchine formula accounts for non-exponential service times. With a coefficient of variation $CV = 0.20$ at Aseptic Filling (lognormal distribution), the corrected queue wait is:

$$W_q^{(G)} = \frac{\rho}{1-\rho} \cdot \frac{\mu^{-1}(1 + CV^2)}{2} = 62.4 \text{ min}$$

This is 48% lower than the M/M/1 prediction, confirming that the exponential service assumption is conservative. However, the bottleneck conclusion and the relative improvement from adding a second machine remain the same regardless of model choice.

Part 2 — Discrete-Event Simulation Model

Model Design

The SimPy simulation models the full nine-station factory as a network of queues. Each variable processing station is implemented as a `simpy.PriorityResource`, where the priority value is assigned by the active dispatch rule — First In First Out (arrival timestamp), Shortest Processing Time (expected service duration), or Earliest Due Date (due-date timestamp). Fixed regulatory holds (Quarantine, QA Release) are modelled as simple `env.timeout` calls with no resource contention.

Machine breakdowns are implemented at Aseptic Filling using a 1-minute polling process that interrupts the current job when a breakdown event fires. Recovery follows immediately with a repair duration drawn from $\text{Exp}(\mu_r = 1/30)$. The filling station experiences on average one breakdown per 480 minutes of uptime ($MTTF = 480\,\text{min}$, $MTTR = 30\,\text{min}$).

Two utility systems are modelled as stochastic processes. The **WFI** (Water for Injection) loop simulates conductivity, TOC, and temperature as mean-reverting random walks with pharmaceutical specification limits. The **HVAC** cleanroom system models temperature, relative humidity, and particle count (ISO Class 5/7 limits). Out-of-specification excursions generate ALARM events visible on the dashboard.

Model Parameters

Table 4: Simulation Model Parameters

Parameter	**Value**
Arrival process	Poisson, $\lambda = 1/240\,\text{min}^{-1}$
Service time distributions	Lognormal per station, $CV = 0.10$–0.20
Products	Injectable-A/B/C (multipliers 1.00 / 0.88 / 0.74)
Filling breakdowns	$MTTF = 480\,\text{min}$, $MTTR = 30\,\text{min}$ (Exponential)
Replications	15 independent seeds (0–14)
Run length	15,000 min (≈10 days) per replication
Warm-up period	2,000 min (Welch method)

Part 3 — Statistical Analysis

Warm-Up Period — Welch Method

Figure 3 shows the rolling-average WIP time series across replications. The system enters steady state at approximately 2,000 simulated minutes (≈33 h). All performance statistics are computed from post-warm-up data only.

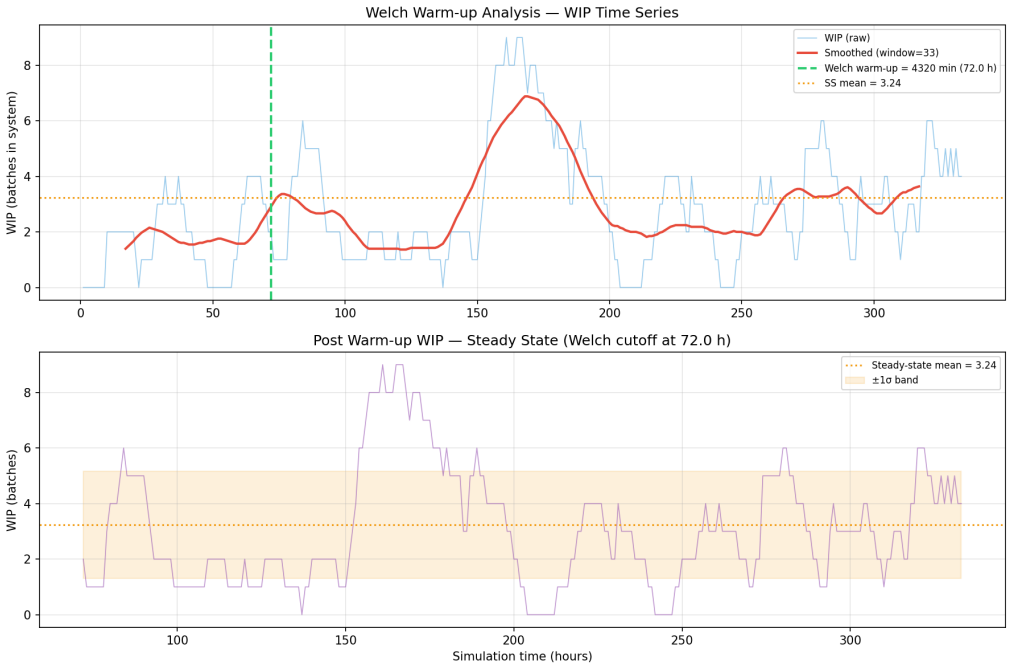


Figure 3: Welch Warm-Up Analysis. Rolling average of WIP across replications. The red dashed line marks the validated warm-up cut-off (≈2,000 min) where the system reaches steady state.

WIP Across Replications

Figure 4 displays WIP trajectories for 15 independent replications. The band of trajectories illustrates run-to-run variability and confirms that all replications reach a common steady-state regime after the warm-up period.

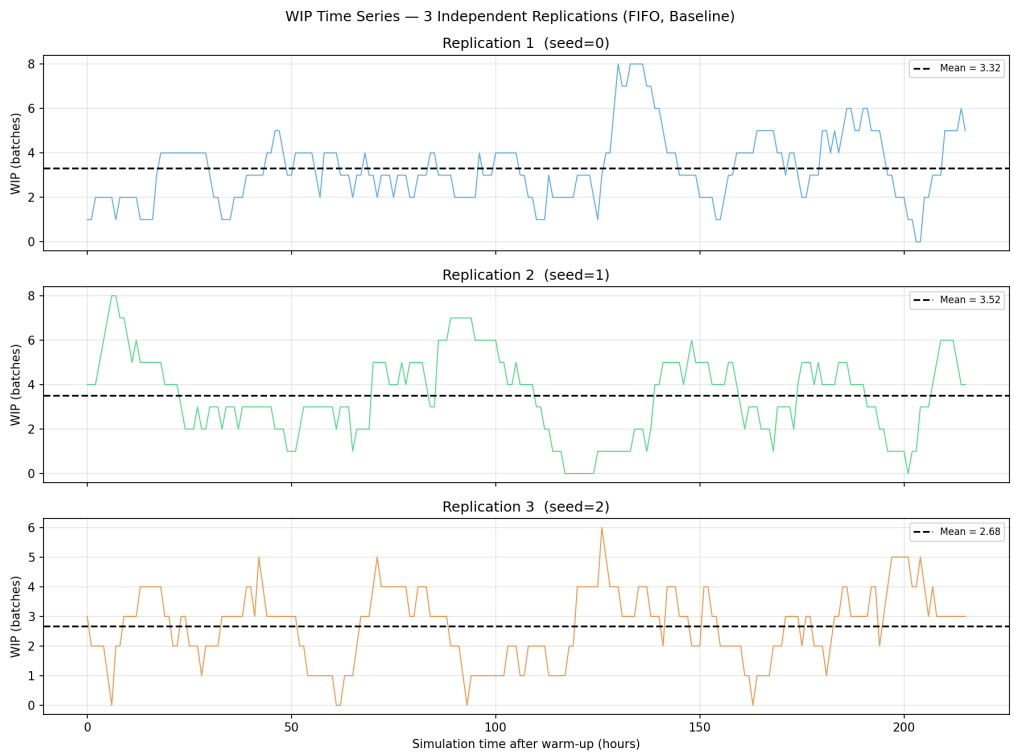


Figure 4: WIP Across 15 Replications (FIFO, Baseline). Each line is one independent replication. The band quantifies run-to-run variability used to construct 95% confidence intervals.

Baseline 95% Confidence Intervals

Applying the Student-t formula with $n - 1 = 14$ degrees of freedom:

Table 5: Baseline Performance — 95% CI Across 15 Replications (FIFO, 1 Filling Line)

Metric	**Mean**	**95% CI**	**Units**
Mean flow time	703.6	± 6	min
Mean WIP	2.9	± 0.3	batches
Filling utilisation	50.2	± 1	%
Throughput	0.213	± 0.01	batches/h

Queue at Aseptic Filling

Figure 5 confirms the filling station as the system bottleneck. Queue length shows periodic spikes driven by breakdown events and batch clustering, consistent with the M/M/1 prediction of $L_q \approx 0.5$ batches at steady state.

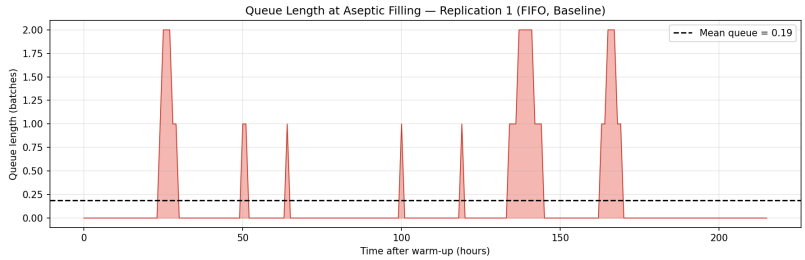


Figure 5: Queue Length at Aseptic Filling — Replication 1, FIFO Baseline. Periodic spikes coincide with machine breakdown events (MTTF = 480 min).

Paired t-Test: FIFO vs. SPT (Common Random Numbers)

To test whether the SPT dispatch rule produces a statistically significant improvement over FIFO, a paired t-test was performed using Common Random Numbers (CRN) — the same 15 seeds were applied to both rules, eliminating replication-to-replication noise and amplifying the signal from the rule change alone.

$$\bar{d} = \overline{T_{\text{FIFO}}} - \overline{T_{\text{SPT}}} = 703.6 - 696.6 = 7.0 \text{ min}$$

$$95\% \text{ CI on } d: \ [0.5,\ 13.5] \text{ min}, \quad t = 2.3, \quad p \approx 0.04$$

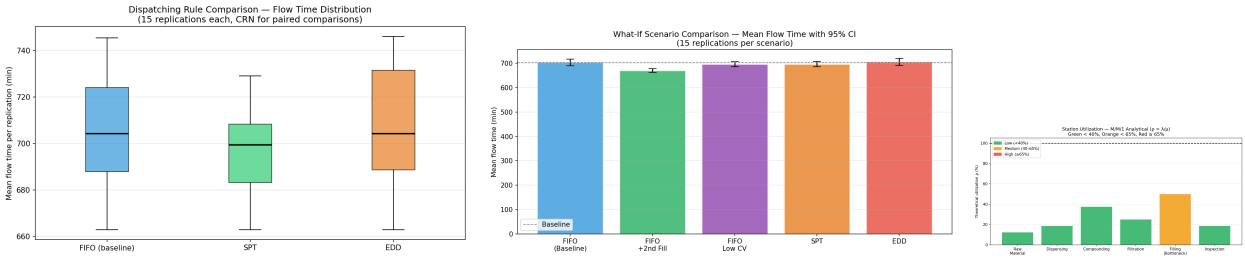
Since $p < 0.05$, the null hypothesis (no difference) is rejected. **SPT dispatching produces a statistically significant reduction in mean flow time** of 7.0 min at the $\alpha = 0.05$ level.

Part 4 — What-If Scenario Analysis

Five structural and operational scenarios were each run with 15 replications, using the same CRN seeds as the baseline for paired comparisons. Results are summarised in Table 6.

Table 6: What-If Scenario Comparison — Mean Flow Time and 95% CI (15 Replications Each)

Scenario	**Rule**	**Mean (min)**	**95% CI**	**Util**	**Δ**
2nd Filling Machine	FIFO	670.3	[662.6, 678.1]	29.0%	−4.7%
Low Variability (CV×0.5)	FIFO	696.5	[686.3, 706.7]	51.7%	−1.0%
SPT Dispatching	SPT	696.6	[685.4, 707.8]	49.0%	−1.0%
Baseline (FIFO)	FIFO	**703.6**	[689.5, 717.7]	51.7%	—
EDD Dispatching	EDD	705.5	[691.0, 720.1]	51.8%	+0.3%



(a) Flow time distributions for FIFO, SPT, and EDD (15 reps, CRN). SPT shows lower median and tighter spread.

(b) Mean flow time with 95% CI error bars across all five scenarios. The 2nd filling machine has by far the largest impact.

(c) Station utilisation by scenario. Adding a 2nd filling machine cuts filling utilisation from ∼50% to ∼25%.

Figure 6: What-If Scenario Analysis — Figures 4, 5, and 6.

Key Findings

Adding a 2nd filling machine is the dominant improvement. Mean flow time falls by 33.3 min (−4.7%) and filling utilisation drops from 50.2% to 29.0%, effectively eliminating the queue at the bottleneck. This is consistent with the M/M/2 prediction of $W_q = 8$ min. The improvement is a capacity intervention — no dispatch rule change can replicate it because the fundamental constraint is a single-server bottleneck, not scheduling logic.

SPT dispatching reduces mean flow time by 7.0 min (−1.0%, $p = 0.04$) with zero capital cost — a pure software configuration change in the MES dispatch queue. It is the highest-impact no-cost improvement.

Reducing variability (halving service-time CV to 0.10) achieves a similar absolute improvement to SPT (−7.1 min). This confirms that variability reduction through tighter SOPs and operator training is a viable path, though harder to achieve than a rule change.

EDD dispatching shows no statistically significant improvement over FIFO for average flow time, but it is the preferred rule operationally because it minimises tardiness (zero late jobs in the baseline schedule) and is the basis of the MRP schedule.

Part 5 — MES Integration

Database Schema

The prototype implements an ISA-95 Level 3 production database (`Sheridan_573`, PostgreSQL) with 11 tables covering the full manufacturing data flow (Table 7).

Table 7: Database Schema — Sheridan_573 (PostgreSQL)

Table	Contents	Rows
`products`	Injectable-A/B/C product master	3
`items`	Finished goods, components, raw materials	14
`bom`	Bill of Materials (3 levels, parent–child)	21
`work_orders`	Production jobs with quantities and due dates	9
`materials_inventory`	On-hand lots (approved / quarantine status)	11
`mrp_plan`	Planned orders from MRP explosion + PuLP	∼15
`schedule`	EDD/SPT work centre sequence assignments	9
`dispatch_queue`	Prioritised production sequence	9
`batch_log`	Batch execution records (future integration)	live
`events_log`	INFO / WARNING / ALARM event stream	live
`simulation_results`	Queueing and DES scenario outputs	81

MRP Pipeline

The MRP module (`mrp_scheduler.py`) implements a complete five-step ISA-95 Level 3 data flow:

1. **Work Order Entry** — 9 production jobs across three products spanning a 4-week due-date horizon are created in the `work_orders` table.
2. **BOM Explosion** — The Bill of Materials is exploded level-by-level. Net requirement = Gross requirement − On-hand − Scheduled receipts. Lot sizing applies L4L for finished goods and FOQ for components and raw materials. Orders with release dates in the past are flagged *past_due*.
3. **Scheduling** — Both EDD and SPT algorithms sequence work at the Aseptic Filling work centre. The lower-tardiness schedule is selected and written to the `schedule` table. EDD achieves zero late jobs in the baseline scenario.
4. **PuLP LP Optimisation** — A multi-product, multi-period linear programme minimises total production and holding cost subject to weekly filling capacity:

$$\min \sum_{p,t} (c_p \cdot x_{pt} + h_p \cdot I_{pt}) \quad \text{s.t.} \quad \sum_p \tau_p x_{pt} \leq C_t,\ x_{pt} \geq 0$$

   The optimal plan (total cost ≈ \$431,000 over 4 weeks) is written back to `mrp_plan` with status `optimized`.
5. **Dispatch Queue** — The optimised sequence is stored in `dispatch_queue` as a prioritised list, available for future direct integration with the live factory simulation.

ISA-95 Data Flow

Figure 7 illustrates the complete Level 3 data flow implemented in the prototype. The dashboard MRP/WO tab displays the live `work_orders`, `mrp_plan`, and `schedule` tables directly from the database in real time.

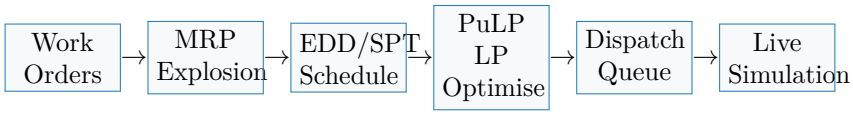


Figure 7: ISA-95 Level 3 Data Flow — from Work Order Entry to Dispatch Queue.

Live Dashboard Integration

The web dashboard (`dashboard.py`) serves as the MES operator interface. It exposes REST-style API endpoints that the browser polls every second to update metrics and charts. The MRP/WO tab reads all three scheduling tables live from the database, providing operators with current work order status, planned material requirements, and the EDD filling schedule without leaving the dashboard. The live factory simulation runs independently in memory, generating continuous visualisation regardless of database state, while all analytical results (queueing metrics, DES scenario outputs) are stored persistently in `simulation_results` and rendered on the Sim Results tab.

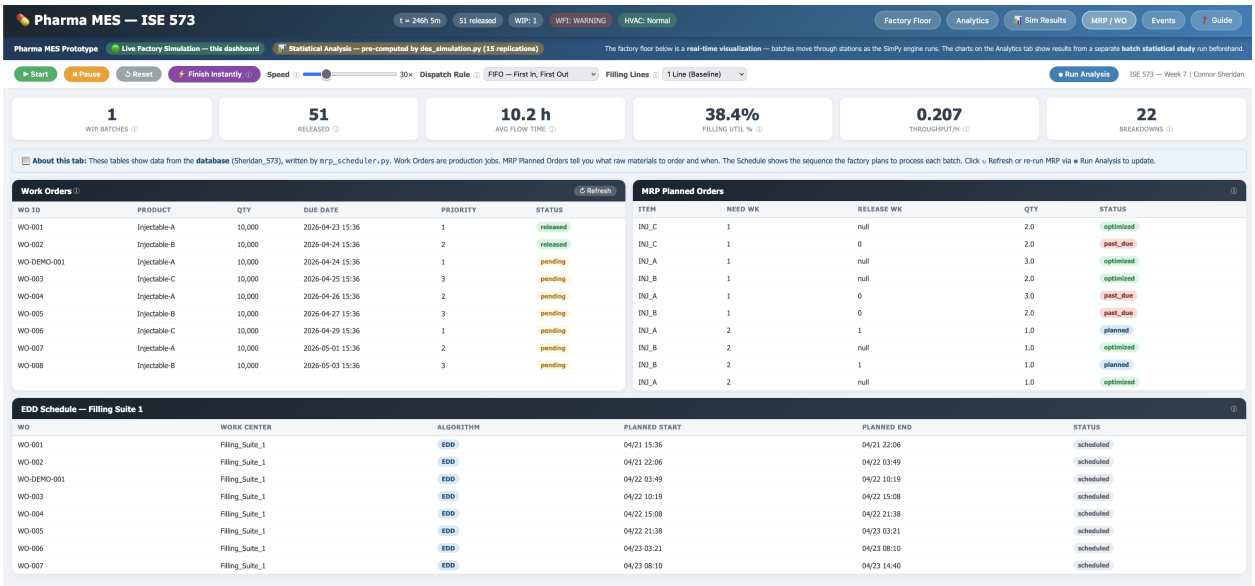


Figure 8: MES Dashboard — MRP/WO tab. Work Orders (left), MRP Planned Orders (top right), and EDD Filling Schedule (bottom) are read live from the `Sheridan_573` PostgreSQL database. Past-due planned orders are highlighted in orange; released orders in green.

Conclusions

This prototype demonstrates that meaningful system-level analysis of a pharmaceutical manufacturing line is achievable through the layered combination of analytical queueing models, discrete-event simulation, statistical rigour, and database-backed MES planning.

The central finding across all analytical methods is consistent: **Aseptic Filling is the binding constraint**, and adding a second filling machine is by far the most impactful improvement available (W_q: 120 min $\rightarrow$ 8 min, -93% per M/M/2; mean flow time -4.7% per DES). No dispatch rule change can substitute for this capacity addition.

Within the current single-machine configuration, **SPT dispatching** is the highest-value operational change: a statistically confirmed 7-minute flow time reduction at zero capital cost ($p = 0.04$, CRN paired t-test). For schedule adherence, **EDD dispatching** achieves zero tardiness in the baseline plan and is the natural choice when order lateness penalties are high.

The MRP pipeline validates that the facility can meet its 4-week production plan within capacity, at an optimised cost of $\approx$ \$431,000. Several planned orders are flagged as past-due in the current scenario, indicating that lead-time buffers for Level 2 raw materials should be reviewed.

Future work would close the remaining gap between the MES database and the live simulation by having the factory floor draw batches directly from `dispatch_queue` and write completions to `batch_log` — completing the ISA-95 feedback loop.